

Stock Intelligence System

A multi-agent AI platform that analyzes any NSE / BSE / US stock across news, technicals, fundamentals, sentiment, and risk — then produces a transparent Buy / Hold / Sell verdict.

BUILT WITH TYPESCRIPT · REACT · GEMINI · POSTGRES

Prepared for	Clients & Investors
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Document version	v1.0

Executive Summary

Stock Intelligence is a full-stack AI platform that turns retail investment research from a multi-hour, multi-tab chore into a single click. The user types any company name — *TCS, Reliance, Apple, Waaree Energies* — and five specialist AI agents go to work in parallel, analyzing news, technical indicators, fundamentals, social sentiment, and risk. A sixth agent synthesizes their outputs into a single **Buy / Hold / Sell** recommendation with confidence scoring and complete reasoning.

Built end-to-end in TypeScript with a multi-agent LangGraph-style orchestration, Retrieval Augmented Generation over uploaded filings, real-time WebSocket price streaming, paper trading, and a configurable alert engine. Indexes 2,300+ NSE/BSE/US listings with fuzzy search.

“Bloomberg-grade analysis for the price of a Netflix subscription.”

2,375+	Stocks indexed (NSE + US)
5	AI agents per analysis
~2 min	End-to-end recommendation time
12	Production-grade phases shipped

The Problem

Retail stock research is broken. To answer a single question — *should I buy this stock?* — a serious investor today juggles:

- **Moneycontrol / Screener** for fundamentals
- **TradingView** for technicals and charting
- **Twitter, Reddit, Telegram** for sentiment
- **Bloomberg / ET / Mint** for news
- **Tickertape / Trendlyne** for risk metrics
- **ChatGPT** to summarize what they just read

Even a disciplined analyst spends **30–60 minutes per stock** switching tabs, copy-pasting, and mentally reconciling contradictory signals. Most retail investors don't. They make decisions on tips, vibes, and YouTube influencers — and lose money predictably.

The Solution

One search box. One click. One verdict. Stock Intelligence collapses the whole research workflow into a single AI-orchestrated pipeline:

- **Search** any of 2,375+ Indian and US listed companies by name or symbol
- **Five AI agents** analyze the stock in parallel (news, technical, fundamentals, sentiment, risk)
- **A recommendation agent** synthesizes their outputs with confidence scoring and a reflection loop
- **Receive a verdict** — Buy, Hold, or Sell — with full transparent reasoning and a citable trace
- **Continuous monitoring** — real-time price alerts via WebSocket and email, paper-trading portfolio
- **Deep dive** — upload company filings (PDF) and chat with them via Retrieval Augmented Generation

Product Overview

The platform is a single web application with 9 user-facing pages, each powered by a domain-specific subsystem. Everything ties back to one central search bar — the user's entire research workflow.

Dashboard	Live-updating watchlist, recent recommendations, alert feed
Stock Analysis	TradingView-grade candle chart, company profile, one-click full analysis
AI Insights	Recommendation history with per-agent breakdown and reflection traces
Agent Monitor	Replay every AI run; see timing, status, success/failure per agent
Market	Index ETFs (SPY, QQQ, NIFTY, SENSEX) and top movers from your universe
RAG Chat	Upload 10-Ks, annual reports, transcripts; ask questions with inline citations
Sentiment	Reddit crowd mood, topic trends, distinct from professional sentiment
Risk	Volatility, drawdown, beta, leverage radar chart
Alerts	Price-change rules with WebSocket toasts + email + auto re-analysis
Portfolio	Paper trading from \$100k cash, live P&L, full trade history

The Multi-Agent AI Workflow

This is the heart of the product. When the user clicks *Run Analysis* on a stock, this happens:

Stage 1 — Parallel Fanout

Five specialist agents launch **in parallel**, each focused on one dimension. Total wall time = the slowest agent, not the sum. Each agent has the same skeleton: pull real data, ground a structured prompt, call the LLM in JSON mode, validate against a Zod schema, persist the result.

News Agent	Fetches recent headlines from NewsAPI + Finnhub, scores impact and sentiment
Technical Agent	Computes RSI, MACD, Bollinger Bands, SMAs in Python, LLM interprets pattern
Fundamentals Agent	Pulls PE, PEG, ROE, debt/equity; optionally augments with RAG over filings
Sentiment Agent	Reads Reddit crowd discussion, gauges mood and discussion volume
Risk Agent	Volatility, drawdown, beta, leverage — score is inverted (+1 = safest)

Stage 2 — Recommendation Synthesis

A sixth **Recommendation Agent** receives all five outputs (and any errors) and produces a single verdict: Buy, Hold, or Sell — with a confidence score in $[0, 1]$, a 1-line summary, and multi-sentence reasoning. It also lists *contributing signals* — how much weight each agent's output got and why.

Stage 3 — Reflection Loop

If confidence is below 0.40, the agent re-runs with the same data plus a self-correction hint — up to 2 reflections. This is a poor-man's chain-of-thought: either the second pass commits more strongly, or it honestly stays uncertain. Either is useful information.

Stage 4 — Persistence + Replay

Every run — success or failure — is written to the database with per-node timings, raw agent outputs, and the full trace. The Agent Monitor page is a complete replay tool. Crucial for trust: every verdict is auditable, never a black box.

Failure-tolerance is built in: if 1–2 agents fail (e.g., Reddit rate-limited the sentiment agent), the workflow continues. The Recommendation Agent is told which signals are missing, factors that into confidence, and still produces a verdict. The system degrades gracefully — it never just crashes.

Technical Architecture

End-to-end TypeScript stack. One language across frontend, backend, agents, and ORM. Single dependency tree, fully type-safe from database column to React component.

System Topology



Data Flow — Real-time Price Pipeline

Every 15 seconds, a background producer fetches live quotes for the union of all watchlist symbols + all active alert symbols + all WebSocket subscribers. Quotes are written to a Redis Stream (at-least-once delivery, replay buffer) and simultaneously broadcast to connected WebSocket clients. An alert evaluator runs in the same loop: any rule whose direction and threshold match the tick fires immediately — emitting a WebSocket toast, an email, and optionally kicking off a fresh full-stack re-analysis.

Technology Stack

Every choice is deliberate. We picked tools that are mature, type-safe, and have minimal vendor lock-in.

Frontend

React 18	Component model + ecosystem
TypeScript	End-to-end type safety; same language as backend
Vite	Sub-second HMR for developer velocity
Tailwind CSS + shadcn/ui	Utility-first styling; no CSS files to manage
lightweight-charts	TradingView's open-source charting library
TanStack Query	Async server-state management with caching
Zustand	Tiny global state for auth/stream stores

Backend

Fastify 4	High-performance HTTP server, plugin ecosystem
TypeScript (Node 20)	Same language as frontend; ESM modules
Prisma 5	Type-safe ORM with auto-generated TypeScript types
Zod	Runtime validation = single source of truth for types
jsonwebtoken + bcrypt	Stateless JWT auth, standard bearer flow
@fastify/websocket	Persistent WS hub with auto-subscribe
ioredis	Battle-tested Redis client supporting Streams + pub/sub
nodemailer	SMTP delivery for alert emails

Data & Storage

PostgreSQL 16	Transactional store; 11 tables, JSONB for agent outputs
Redis 7	Sub-millisecond cache + Streams for at-least-once delivery
ChromaDB 0.5	Open-source vector store for RAG
Supabase (prod)	Managed Postgres with connection pooling
Upstash (prod)	Managed serverless Redis with TLS

AI / LLM Layer

Google Gemini 2.0	Primary LLM provider; 1500 free requests/day
Provider abstraction	Same code runs against Gemini, OpenAI, or local Ollama
text-embedding-004	Gemini's embedding model for RAG vector indexing
Custom workflow	Hand-rolled DAG (no LangGraph.js dependency)
technicalindicators npm	Pure-JS RSI, MACD, Bollinger, SMA computation
yahoo-finance + Finnhub	Multi-source market data with automatic fallback
NewsAPI	Headline aggregation for the News Agent

Infrastructure & DevOps

Docker Compose	Whole stack with one command for local dev
nginx	Reverse proxy + TLS termination in production
Vercel (free)	Global CDN frontend hosting with GitHub auto-deploy
Render (free)	Backend hosting with auto-deploy from main branch
Cloudflare Tunnel	Optional secure tunnel for laptop demos
GitHub	Source control with the entire repo at SiddharthSivanathan/Stock-intelligence-agent

Standout Features

Universal Stock Search

2,375+ Indian and US stocks indexed from the official NSE EQUITY_L.csv. Fuzzy search by name ("reliance", "tata") or symbol (TCS, AAPL, ^NSEI). Results ranked by exact match > prefix > substring > market cap boost.

Multi-Agent AI Workflow

Five specialist LLM agents run in parallel, then a sixth synthesizes. Self-corrects via reflection loop when initial confidence is low. Every run is fully replayable from the Agent Monitor page.

Retrieval Augmented Generation

Upload a company's 10-K or annual report (PDF). Ask questions in natural language: "What did management say about margin pressure?" Get back an answer with inline citations to the exact source chunks.

Real-time Price Streaming

WebSocket hub auto-subscribes each user to their watchlist symbols. Background producer ticks every 15 seconds, broadcasts via WS, and evaluates alert rules in the same loop.

Configurable Alert Engine

Define rules like "AAPL moves above +2%, cooldown 1 hour." Choose WebSocket toast, email, or auto re-analysis. Producer auto-includes alert symbols even when not in watchlist — no silent dormancy.

Paper Trading Portfolio

\$100,000 virtual cash, buy/sell at live quotes, real-time P&L; computation with cost-basis tracking, full trade history. Reset anytime. Test the AI's recommendations risk-free.

Multi-Provider LLM Abstraction

Same code runs against Google Gemini, OpenAI, or local Ollama Mistral. Switched by a single environment variable. No vendor lock-in.

Production-Grade Audit Trail

Every agent run writes a row with input, output, timing, status, model, and provider. Crucial for trust: every AI verdict is auditable. Never a black box.

Market Opportunity

India has crossed 150 million demat accounts (NSE data, 2024). The vast majority of these investors have no professional research support. The informal advisory market — YouTube finfluencers, Telegram tipsters, WhatsApp groups — is large, unregulated, and often predatory.

Market Size

TAM	Global retail investor research: ~\$8B annually (Bloomberg + Reuters + Tickertape segment)
SAM	Indian retail investors actively researching: 30M+ users, growing 20% YoY
SOM	Tech-savvy retail in tier-1 cities, willing to pay \$5-10/month: ~3M users by 2027

Competitive Landscape

Tickertape (India)	Strong fundamentals data but no AI synthesis; user must interpret
Trendlyne (India)	Quant scoring but black-box, no transparent reasoning, no chat
TradingView	Best-in-class charting but technicals-only, no fundamentals or AI
Bloomberg Terminal	Gold standard, but \$20,000/year and not accessible to retail
ChatGPT (general)	Has knowledge but no real-time data, no structured output, no audit trail

Our Differentiation

- **Multi-agent transparency.** Every verdict shows its work — which agent contributed what, with what confidence, and why.
- **Indian-first.** NSE/BSE listings from the actual official CSV. Symbol aliases (NIFTY, SENSEX, BankNifty) work out of the box.
- **End-to-end open architecture.** No vendor lock-in — LLM, database, and hosting are all swappable. Self-hostable.
- **RAG over filings.** Upload an annual report; ask questions of it. No competitor in the retail space offers this.

Business Model

A clean three-tier SaaS structure. Cost-to-serve is dominated by LLM tokens; gross margin improves materially when self-hosted on cheap compute with smaller open-source models for routine analyses.

Free	■0/mo	10 analyses/month, 1 watchlist (10 symbols), email alerts only, no RAG
Pro	■499/mo	Unlimited analyses, 5 watchlists, RAG (5 docs), live WS, priority queue
Pro+	■1,999/mo	Everything in Pro + API access + portfolio backtesting + dedicated quota
Enterprise	Custom	White-label, on-prem deployment, custom agents, SLA

Unit Economics (illustrative)

Each full analysis costs roughly ■1–3 in Gemini API tokens (varies by stock data size). At ■499/mo Pro, an average user running 20 analyses/month costs us ■20–60 — gross margin ~88–96%. Self-hosted open-source models drop this to near zero with modest infra cost.

Other Revenue Streams (later)

- **API access** — quants and small hedge funds pay for the endpoint, not the UI
- **Broker partnerships** — embed the analysis engine inside broker apps for affiliate fees
- **Premium data resale** — offer paid-tier data (Bloomberg, Reuters) as add-on
- **White-label** — rebrand for advisory firms, RIAs, family offices

Roadmap

What's shipped (v1.0)

- Multi-agent AI workflow with reflection loop
- 2,375-stock universal NSE/BSE/US master with fuzzy search
- RAG over PDF uploads with inline citations
- Real-time WebSocket price streaming and alerts
- Paper-trading portfolio with live P&L
- Multi-provider LLM abstraction (Gemini, OpenAI, Ollama)
- Production deployment on Vercel + Render free tiers

Next 90 days (Q1)

- **Indian fundamentals depth** — ROCE, promoter holding, institutional holdings via paid data integration (Tickertape API)
- **Mobile app** — React Native, share core TypeScript with web
- **Scheduled background sync** — daily stock master refresh, fundamentals snapshots, news caching
- **User authentication 2.0** — Google OAuth, email verification, password reset
- **Production-grade observability** — Sentry error tracking, structured logging, request tracing

6–12 months

- **Backtesting engine** — replay agents on historical data; measure recommendation hit rate; show users "the AI would have said X six months ago"
- **Custom agents** — let users build their own specialist agents ("ESG agent", "value-investor agent") with no code
- **Multi-asset** — crypto, FX, commodities; the architecture supports it — just needs symbol routing
- **Community layer** — share recommendation traces, vote on verdicts, follow other users' watchlists
- **API tier** — public REST endpoints for the analysis engine; billed per call

Long term (12+ months)

- **Auto-trading integration** — connect to Zerodha Kite, Upstox; execute the AI's verdict with one tap (with safeguards)
- **Fine-tuned LLM** — train a small specialist model on our captured agent traces; drop inference cost by 10x
- **Enterprise white-label** — advisory firms rebrand and deploy on their own infra

Why Now

Three trends are converging that make this product inevitable today, where it wasn't even two years ago:

- **LLMs hit the quality threshold** for structured financial reasoning. Gemini 2.0 and GPT-4 can read filings and produce audit-grade analysis at *milliseconds-per-dollar* cost.
- **India retail investor base 3x'd** in five years — 50M to 150M demat accounts. Most without research support.
- **Open-source ML tooling matured.** LangGraph, Prisma, Zod, shadcn/ui — the velocity to build a production system like this is now weeks, not years.

Ask

We're seeking partners — advisors, design partners, early customers, and capital — to take this from a working production platform to a category-defining business.

Demo & Repo

Live demo	https://stock-intelligence-agent.vercel.app
Source code	github.com/SiddharthSivanathan/Stock-intelligence-agent
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“The best AI products feel like they read your mind. Stock Intelligence reads the market’s.”